

DATABASE MANAGEMENT SYSTEM

Unit-II: File Organization & Relational Model

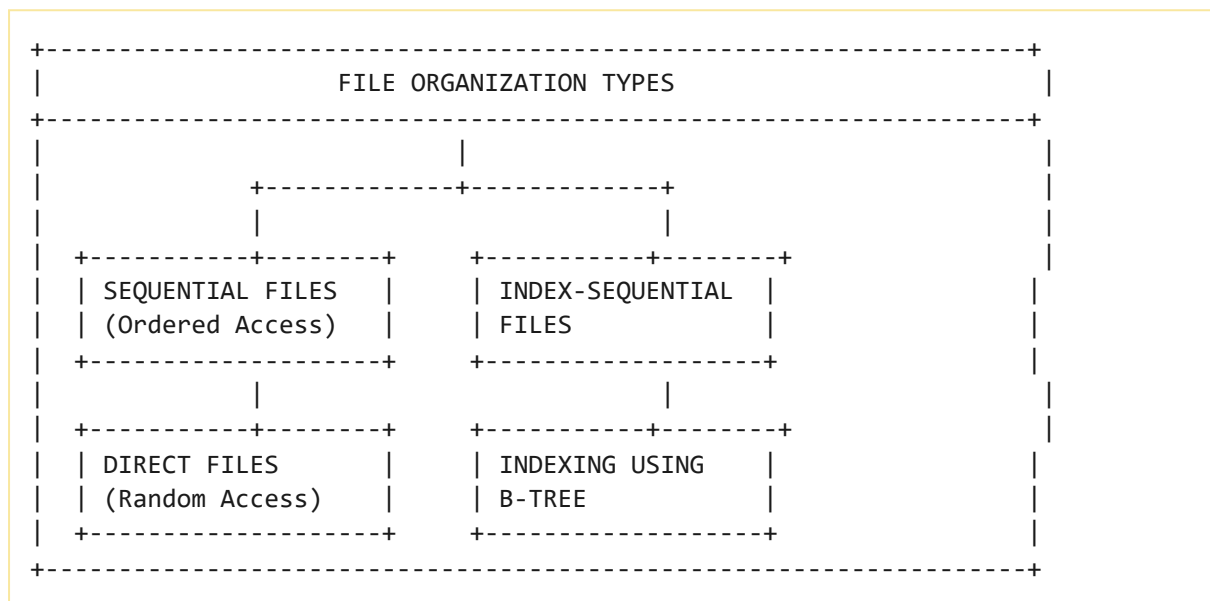
BCA Syllabus - HPU

Comprehensive Study Notes

Topic 1: File Organization

1.1 What is File Organization?

File organization refers to how data is stored and arranged on storage devices (like hard disks) to enable efficient access, retrieval, and modification of records.



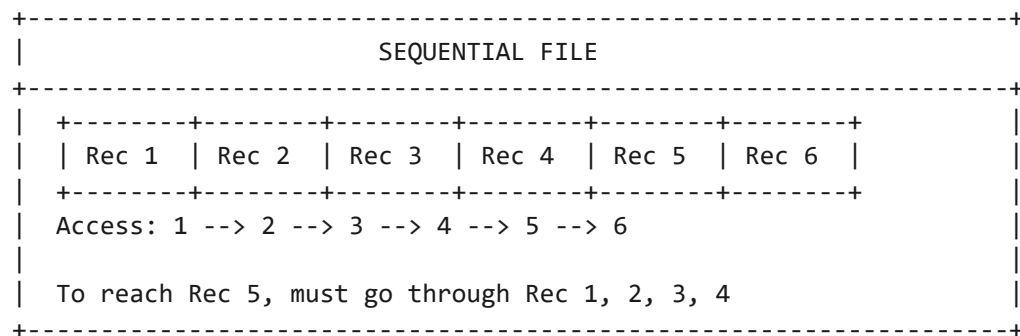
1.2 Operation on Files

Operation	Description
Insertion	Adding a new record to the file
Deletion	Removing an existing record from the file
Updation	Modifying an existing record
Retrieval	Reading records from the file
Search	Finding a record based on a key
Traversal	Accessing all records one by one

Topic 2: Sequential Files

2.1 Definition

A **sequential file** stores records in a linear sequence, typically in the order they were entered. Records are accessed one after another from the beginning to the end.



2.2 Characteristics of Sequential Files

Feature	Description
Order	Records are stored in a specific order (typically by key)
Access	Sequential access only
Insertion	Requires rewriting the entire file
Deletion	Requires marking as deleted or rewriting
Search	Linear search ($O(n)$)
Performance	Good for batch processing
Example	Magnetic tape storage

2.3 Advantages and Disadvantages

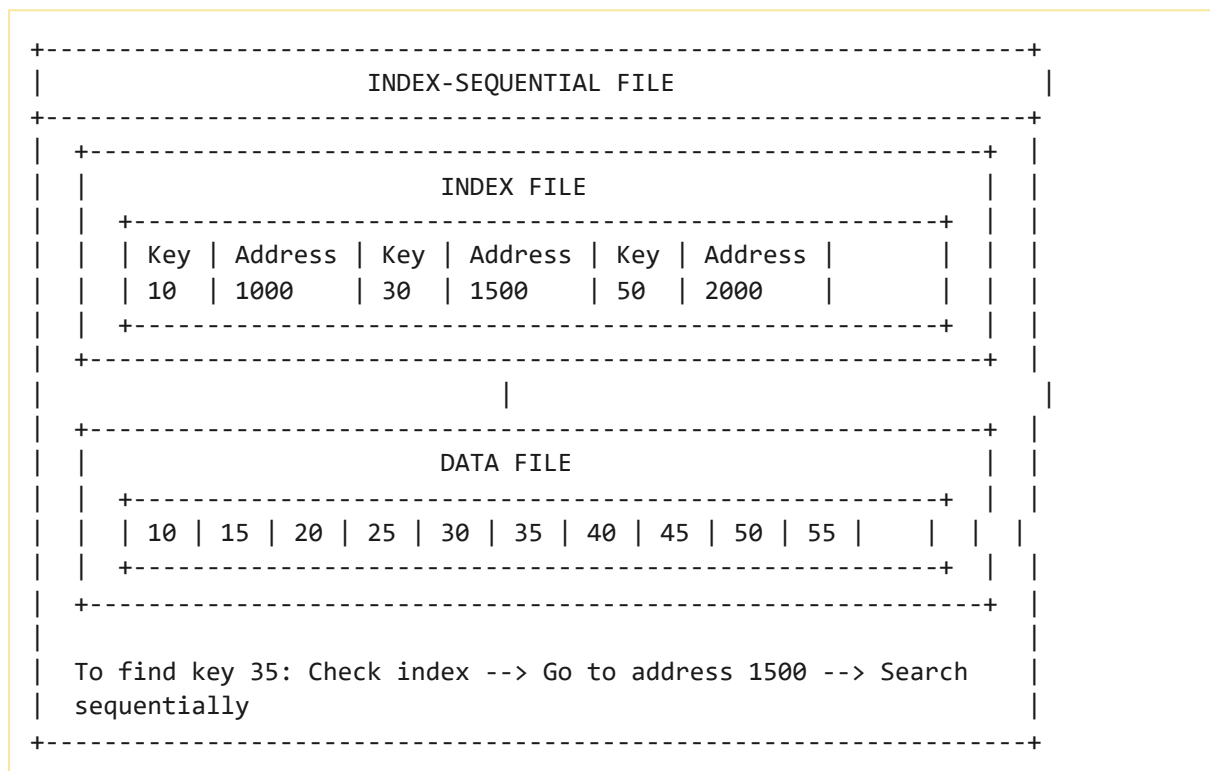
Advantages**Disadvantages**

Simple to implement	Slow access time for random retrieval
No additional storage overhead	Insertion and deletion are expensive
Good for batch processing	Not suitable for interactive applications
Works well with tape drives	Need to search entire file sequentially

Topic 3: Index-Sequential Files

3.1 Definition

Index-sequential files combine both sequential and direct access methods. They maintain an **index** that allows quick access to records while keeping the sequential order.



3.2 Types of Indexes

Index Type	Description	Characteristics
Primary Index	Ordered on primary key	One-to-one mapping
Secondary Index	Ordered on non-primary key field	Many-to-one mapping
Clustering Index	On ordered non-key field	One-to-many mapping
Unique Index	Ensures unique values	No duplicates allowed

3.3 Implicit Index

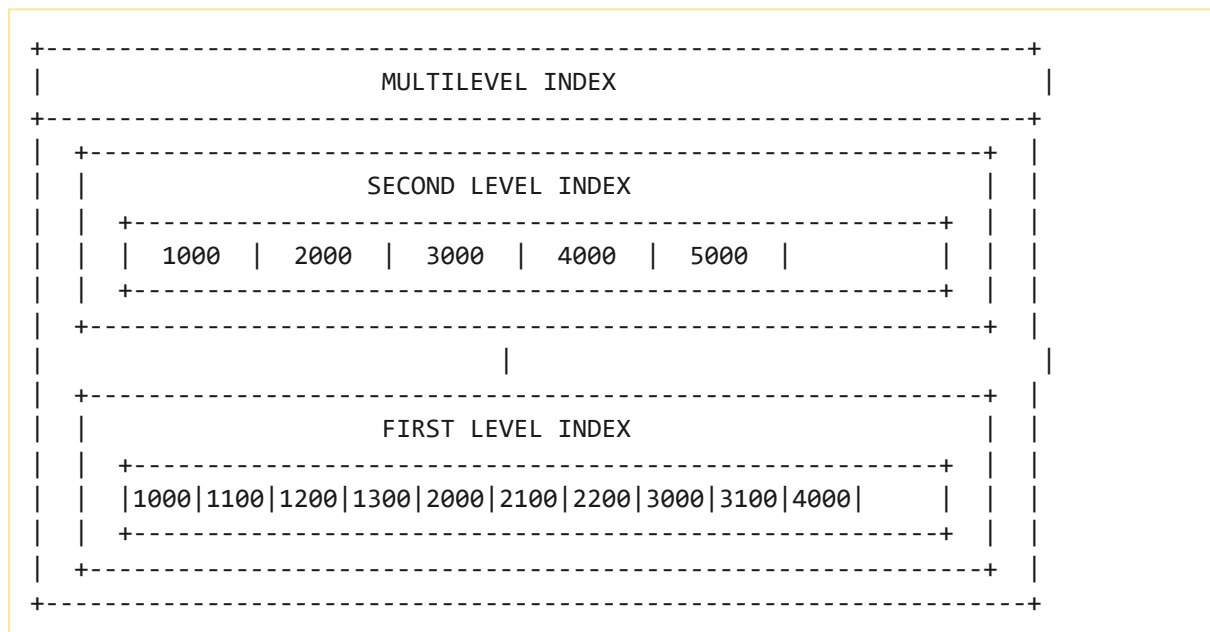
An **implicit index** is automatically created by the DBMS, usually on primary keys, to enforce uniqueness and speed up queries.

3.4 Limit Index

A **limit index** restricts the number of entries in the index to a specific size, improving search efficiency.

3.5 Multilevel Index

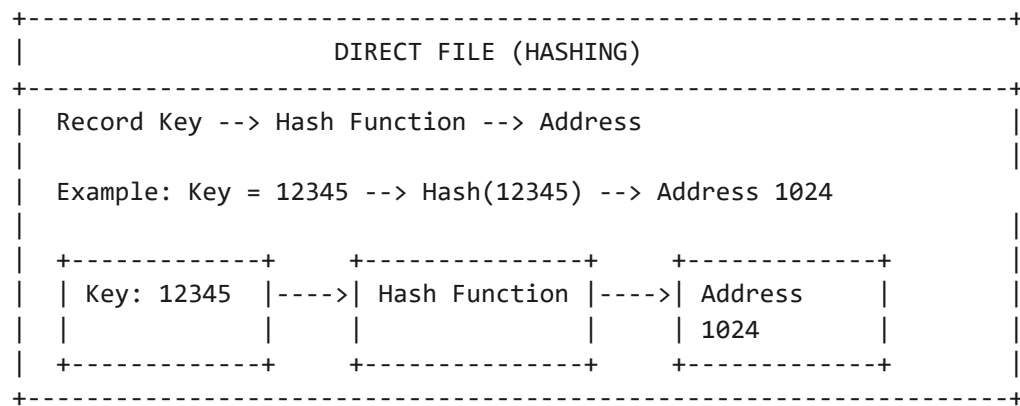
Multilevel indexing is used when the index becomes too large to fit in memory. A second-level index is created for the first-level index.



Topic 4: Direct Files

4.1 Definition

Direct files (also called hash files) use a **hashing function** to calculate the physical location of a record from its key value.



4.2 Characteristics of Direct Files

Feature	Description
Access	Direct access to any record
Speed	Very fast retrieval ($O(1)$ average)
Storage	May have unused space
Collision	Two keys may map to same address
Collision Handling	Linear probing, chaining

4.3 Advantages and Disadvantages

Advantages

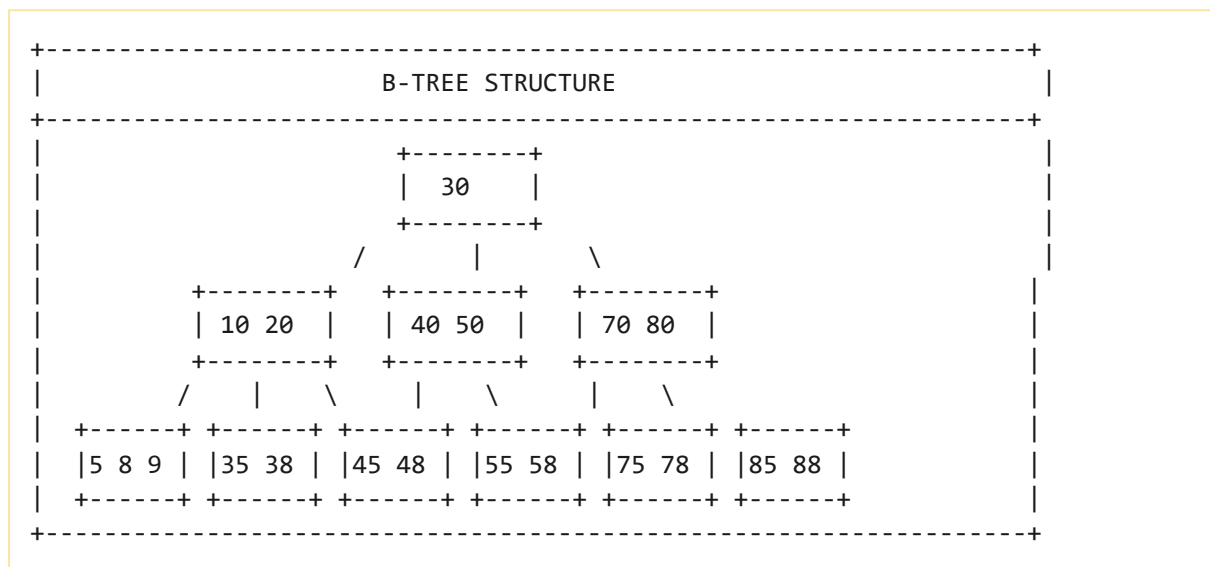
Disadvantages

Fast random access	Wasted storage space
Efficient for large datasets	Collision handling overhead
Simple to implement	Not good for sequential processing

Topic 5: Indexing Using B-Tree Structure

5.1 What is a B-Tree?

A **B-Tree** is a balanced tree data structure that maintains sorted data and allows efficient insertion, deletion, and search operations.



5.2 Properties of B-Tree

Property	Description
Balanced	All leaf nodes are at the same level
Order m	Each node can have at most m children
Sorted	Keys in each node are sorted
Search	$O(\log n)$ search time
Insert/Delete	$O(\log n)$ time
Utilization	At least half full

5.3 B-Tree vs B+ Tree

Feature	B-Tree	B+ Tree
Data Storage	Keys and data in all nodes	Data only in leaf nodes
Search Time	$O(\log n)$	$O(\log n)$
Range Queries	Less efficient	Very efficient
Index Size	Larger	Smaller
Used In	General indexing	Database indexing

Topic 6: Relational Model

6.1 What is the Relational Model?

The **Relational Model** was proposed by **E.F. Codd** in 1970. It represents data as a collection of **relations (tables)** with rows and columns.

RELATIONAL DATABASE			
STUDENT		COURSE	
StudentID	Name	CourseID	CName
S001	Rahul	C101	DBMS
S002	Priya	C102	Python
S003	Amit	C103	Java
ENROLLMENT			
StudentID	CourseID		
S001	C101		
S002	C102		
S003	C103		

6.2 Key Concepts of Relational Model

Concept	Description
Relation (Table)	A two-dimensional table with rows and columns
Attribute (Column)	A named column in a table

Tuple (Row)	A single row in a table
Schema	Structure of the database (table names and attributes)
Instance	The actual data in the database at a given time
Domain	Set of allowed values for an attribute

6.3 Keys in Relational Model

Key Type	Description
Super Key	A set of attributes that uniquely identifies a tuple
Candidate Key	Minimal super key
Primary Key	Chosen candidate key
Alternate Key	Candidate key not chosen as primary
Foreign Key	Key used to link two tables
Composite Key	Key made up of multiple attributes

Topic 7: Relational Algebra

7.1 What is Relational Algebra?

Relational Algebra is a procedural query language that consists of a set of operations that take relations as input and produce relations as output.

Fundamental Operations:

Operation	Symbol	Description
Select	σ (Sigma)	Selects rows based on condition
Project	π (Pi)	Selects specific columns
Union	$\cup$	Combines two relations
Set Difference	$-$	Removes common rows
Cartesian Product	$\times$	Combines all rows from two relations
Rename	ρ (Rho)	Renames attributes

Additional Operations:

Operation	Description
Join	Combines tables based on common attributes
Division	Finds rows in one table that match all rows in another
Intersection	Finds common rows in two relations

7.2 Relational Algebra Examples

Query: Find names of all students.

π Name (STUDENT)

Query: Find students with age > 20.

σ Age > 20 (STUDENT)

Query: Find names of students enrolled in C101.

π Name (STUDENT $\bowtie$ ENROLLMENT) where CourseID = 'C101'

Topic 8: Relational Calculus

8.1 What is Relational Calculus?

Relational Calculus is a non-procedural query language that specifies what to retrieve, not how to retrieve it.

Types:

Type	Description
Tuple Relational Calculus (TRC)	Queries based on tuples
Domain Relational Calculus (DRC)	Queries based on domains

TRC Example:

```
{t | t ∈ STUDENT ∧ t.Age > 20}
```

(Finds all student tuples where age > 20)

8.2 Relational Algebra vs Relational Calculus

Feature	Relational Algebra	Relational Calculus
Nature	Procedural	Non-procedural
Focus	How to get results	What to get
Operations	Operators (σ , π , $\cup$)	Predicates
Expression	Formula	Predicate calculus
Ease	More complex	More user-friendly

Topic 9: HPU Important Questions

Short Answer Questions (2-5 Marks)

1. What is file organization?
2. What is a sequential file?
3. What is an index-sequential file?
4. What are the types of indexes?
5. What is a direct file?
6. What is a B-Tree?
7. What is the relational model?
8. What is a primary key?
9. What is a foreign key?
10. What is relational algebra?

Long Answer Questions (10 Marks)

1. Explain the types of file organization with their advantages and disadvantages.
2. Explain B-Tree indexing with a diagram.
3. Explain the relational model with keys.
4. Explain relational algebra operations with examples.
5. Differentiate between B-Tree and B+ Tree.
6. Explain relational calculus with examples.

Objective Questions (1 Mark)

1. Sequential files use:
(a) Direct access (b) Sequential access (c) Both (d) None
Answer: (b) Sequential access
2. A B-Tree is a:
(a) Linear structure (b) Tree structure (c) Network structure (d) None
Answer: (b) Tree structure
3. Primary key is:
(a) Unique (b) Not null (c) Both (d) None

Answer: (c) Both

4. Foreign key references:

(a) Same table (b) Another table (c) Both (d) None

Answer: (b) Another table

5. Relational algebra is:

(a) Procedural (b) Non-procedural (c) Both (d) None

Answer: (a) Procedural

Quick Revision

File Organization Summary

Type	Access	Speed	Use Case
Sequential	Sequential	Slow for random	Batch processing
Index-Sequential	Sequential + Direct	Medium	Balanced access
Direct	Direct	Fast	Random access
B-Tree	Balanced	$O(\log n)$	Database indexing

Keys Summary

Key Type	Description
Primary Key	Unique identifier
Foreign Key	Links two tables
Candidate Key	Potential primary keys
Composite Key	Multiple attributes

Relational Algebra Operations

Operation	Symbol	Purpose
-----------	--------	---------

Select	σ	Filter rows
Project	π	Filter columns
Union	$\cup$	Combine
Join	$\bowtie$	Combine tables

End of Unit-II Notes

DATABASE MANAGEMENT SYSTEM - BCA Syllabus (HPU)